

scGPT Human Validation: Category-Specific Precision and Recall

Provisional results and re-review recommendation

Purpose

This document summarizes category-specific precision and recall for the 40-study classified validation sample, using the two-reviewer human consensus as the reference standard. The separate 10-study abstention and process-audit sample is not included in these classification metrics.

Each use category is evaluated independently in a one-vs-rest framework. A study may be positive for more than one use category. Unweighted precision describes the reviewed sample; inverse-probability-weighted (IPW) precision estimates performance in the corrected 400-study classified population.

Category-specific results

Counts are shown as human-reference positives (H+), true positives (TP), false positives (FP), and false negatives (FN). “N/E” means not estimable because no human or agent positives were observed.

Category	H+ / TP / FP / FN	Precision Sample / IPW	Recall Sample / IPW
Apply to biological data	5 / 5 / 4 / 0	0.556 / 0.521	1.000 / 1.000
Extend or develop	6 / 6 / 9 / 0	0.400 / 0.380	1.000 / 1.000
Benchmark or evaluate	11 / 11 / 6 / 0	0.647 / 0.804	1.000 / 1.000
Mention only	20 / 20 / 0 / 0	1.000 / 1.000	1.000 / 1.000
Other executed use	0 / 0 / 0 / 0	N/E / N/E	N/E / N/E

Interpretation

The agent recovered every human-positive category in this sample (no false negatives), but it assigned additional Apply, Extend, and Benchmark labels. Over-labeling was greatest for Extend or develop. The recall values of 1.000 are point estimates based on small category counts and require uncertainty intervals before manuscript reporting.

These values are precision and recall at the agent's fixed classification rule, not precision–recall curves or area under the precision–recall curve (AUPRC). A curve requires a prespecified continuous score for each category; the current hard labels alone cannot provide one.

Why the first-round labels should be revisited

The first-round review form asked for a singular “Primary use” and presented the use categories as a single list. That wording may have encouraged reviewers to select one category even when a paper served more than one purpose. By contrast, the agent contract used non-exclusive use_labels. After reconciliation, HV-022 and HV-049 were recorded as both Apply to biological data and Extend or develop, suggesting that additional multi-purpose uses may remain under-annotated.

Recommended blinded re-review

Before these category-level metrics are used in a manuscript, both reviewers should independently revisit all 20 studies in the classified validation subset for which scGPT was judged executed. The revised question should read: “Select all applicable uses of scGPT: Apply to biological data; Extend or develop; Benchmark or evaluate; Other executed use.” A separate primary-use field may be completed afterward for descriptive summaries only.

Reviewers should remain blinded to all agent labels, confidence, and evidence selections during this revisit. Their revised multi-label records should then be reconciled, frozen, and used to rerun the one-vs-rest precision, recall, F1, exact-set agreement, and Jaccard calculations. The three human-executed studies in the separate abstention/process audit may also be revisited, but their results should remain outside the 40-study classification-accuracy denominator.

Reporting status

All values in the table are provisional. They should not be treated as manuscript-final category performance until the multi-select re-review is completed.